Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI for Cyber Security

Course Code: C434

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	4 October 2026	Initial release of AI for Cyber Security (C434): two-day 9:30am–5:30pm lesson plan covering all four topics and twelve labs.	Dr Alfred Ang

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Course Overview

AI for Cyber Security is a two-day, 15-hour course (9:30am – 5:30pm daily). It combines 7 classroom hours and 8 hands-on practical hours.

Learning Outcomes

- LO1: Identify security risks, threats, and assess gaps in controls impacting business processes.
- LO2: Translate security objectives into specific programmes considering security programme design techniques.
- LO3: Develop a detailed action plan for security programmes in line with information security architectures.
- LO4: Deliver advice for adoption of security architectures, monitoring their effectiveness against standards.

Daily Schedule

Day 1 • 9:30am – 5:30pm: Risk, detection and AI security programme design (Topics 1–2)

Time	Duration	Topic / Activity	Method	Slides
9:30-9:45	15 min	Welcome, introductions, ground rules and course orientation	Classroom	Slides 1-11
9:45-11:15	90 min	Topic 1 mechanism, contracts, failure, control and evidence	Classroom	Slides 12-101
11:15-12:30	75 min	Labs 1-2 guided practice	Practical	Slides 102-105
12:30-13:00	30 min	Lunch	Break	-
13:00-13:45	45 min	Labs 3-4 guided practice and evidence review	Practical	Slides 106-109
13:45-15:15	90 min	Topic 2 mechanism, contracts, failure, control and evidence	Classroom	Slides 110-199
15:15-17:15	120 min	Labs 5-7 guided practice and evidence review	Practical	Slides 200-205
17:15-17:30	15 min	Day 1 recap and Q&A	Classroom	Slides 12-205

Day total: 7.5 instructional hours (9:30am – 5:30pm), with a 30-minute lunch break.

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Day 2 • 9:30am – 5:30pm: Architecture, guarded response, monitoring and governance (Topics 3–4)

Time	Duration	Topic / Activity	Method	Slides
9:30-9:45	15 min	Day 1 recap	Classroom	Slides 12-205
9:45-11:15	90 min	Topic 3 mechanism, contracts, failure, control and evidence	Classroom	Slides 206-294
11:15-12:30	75 min	Labs 8-9 guided practice	Practical	Slides 295-298
12:30-13:00	30 min	Lunch	Break	-
13:00-14:15	75 min	Labs 10-11 guided practice and evidence review	Practical	Slides 299-302
14:15-15:30	75 min	Topic 4 metrics, drift, adversarial testing and standards	Classroom	Slides 303-391
15:30-17:00	90 min	Lab 12 monitoring and advisory report	Practical	Slides 392-393
17:00-17:30	30 min	Governance advice, course synthesis, Q&A and course feedback	Classroom	Slides 394-399

Day total: 7.5 instructional hours (9:30am – 5:30pm), with a 30-minute lunch break.

Topic-by-Topic Breakdown

Topic 1: AI-Assisted Security Risk, Threat and Control-Gap Analysis • Slides 12-109

Alignment: LO1. Telemetry contracts · detection models · business impact · analyst evidence

Lab 1: Build an AI-Assisted Cyber Risk Register • Slides 102-103

Join asset criticality, threat likelihood, control effectiveness and evidence into a transparent residual-risk ranking.

Lab 2: Detect Authentication Anomalies • Slides 104-105

Create a peer-aware anomaly score from failed logins, new-device state and impossible-travel indicators.

Lab 3: Explain a Phishing Risk Score • Slides 106-107

Combine email authentication, reply-to mismatch, URL and language features into an explainable phishing triage score.

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Lab 4: Prioritise Vulnerabilities with Context · Slides 108-109

Rank vulnerabilities using severity, exploit likelihood, exposure, asset criticality and compensating controls.

Topic 2: Designing AI-Powered Cybersecurity Programmes · Slides 110-205

Alignment: LO2. Objectives · use-case selection · governance · human oversight · operating model

Lab 5: Select and Scope an AI Security Use Case · Slides 200-201

Score candidate use cases on value, data readiness, integration effort, operating risk and human-oversight requirements.

Lab 6: Threat-Model an AI SOC Assistant · Slides 202-203

Map assets, trust boundaries and AI-specific abuse cases, then turn them into programme guardrails.

Lab 7: Build the AI Security Programme Charter · Slides 204-205

Translate assessed control gaps into governed programme epics, metrics, owners and decision gates.

Topic 3: Developing AI-Driven Security Architecture and Action Plans · Slides 206-302

Alignment: LO3. SOC architecture · data pipeline · guarded automation · delivery roadmap

Lab 8: Design an AI-Enhanced SOC Architecture · Slides 295-296

Design a layered architecture that separates evidence, inference, case management and guarded action.

Lab 9: Build a Grounded Threat-Intelligence Assistant · Slides 297-298

Retrieve from approved threat bulletins and produce cited, abstaining summaries without external APIs.

Lab 10: Design a Guarded SOAR Playbook · Slides 299-300

Build a reversible, approval-gated response playbook for suspicious account activity.

Lab 11: Create a 90-Day AI Security Action Plan · Slides 301-302

Sequence architecture work packages, dependencies, acceptance evidence and regulatory review triggers.

Topic 4: Monitoring, Evaluating and Governing AI Security Solutions · Slides 303-393

Alignment: LO4. Metrics · drift · adversarial testing · standards · advisory reporting

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Lab 12: Monitor Model Effectiveness and Produce Advice · Slides 392-393

Measure detection quality, identify drift and control failures, then issue evidence-backed adoption advice.

Resources Required

- Windows or macOS laptop with Python 3
- Local virtual environment and lab requirements.txt
- Current browser for the supplied cybersecurity simulations
- Optional isolated trainer-owned Kali VM; never target unrelated live systems
- Course LMS/TMS access
- Individual synthetic and mock-data bundles supplied; no production data required
- Projector and Internet access

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